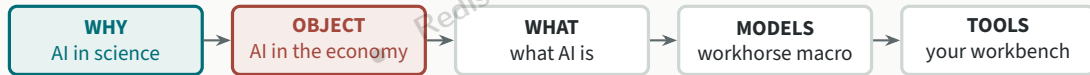


## Session 1.2

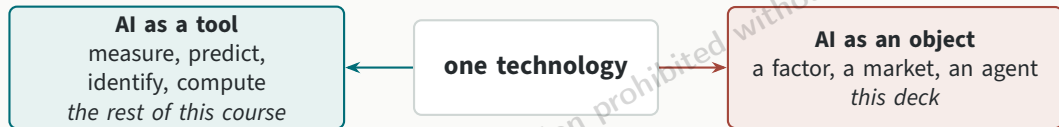
# The Economics of Producing Intelligence

From the facts of Deck 1.1 to research questions: the factor, its unit, its measurement, and its risks



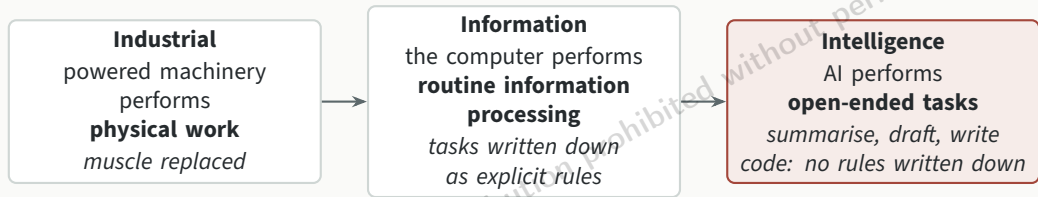
Zhigang Feng

# AI as Tool Versus AI as Object



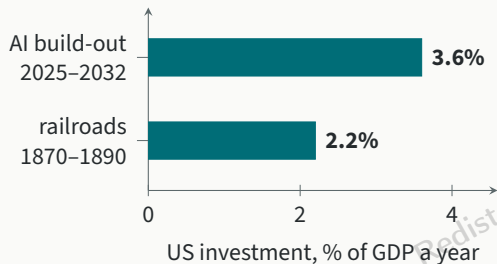
- ▶ Deck 1.1 gave the facts: frontier proofs, agents at scale, a bottleneck moving to verification.
- ▶ This deck asks what they mean for economics, and which questions are still open.
- ▶ **The model you use as an estimator is also the treatment whose effects you may study.**

# Three Revolutions



- ▶ Each wave is defined by the kind of work machines came to perform.
- ▶ The second automated **routine tasks**: work someone could write down as explicit rules and hand to a machine to execute.
- ▶ The third performs tasks for which no one has written the rules. A model learns them from recorded examples.
- ▶ **The third wave's product is cognition itself.**

# Intelligence Is Entering the Economy Explicitly



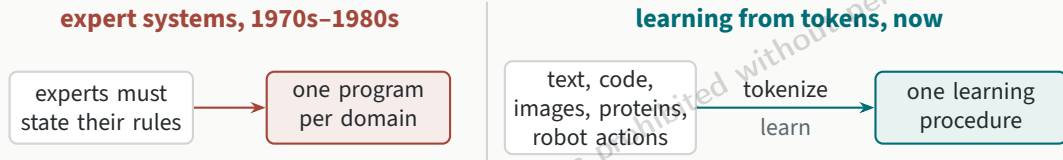
- ▶ Data centres, power, networks and chips: about \$10.3 trillion over 2025–2032, a central scenario built from announced projects, not committed spending.
- ▶ The output is sold by the call, at a posted price.
- ▶ Zuckerberg (Meta, Q2 2026): “a significantly higher margin on selling intelligence rather than selling compute directly”.
- ▶ **Intelligence is now produced, priced and sold.**

## A New Kind of Factor

|               | rival in use                                                     | non-rival in use                                              |
|---------------|------------------------------------------------------------------|---------------------------------------------------------------|
| slow to build | <b>human capital</b><br>a skill built over a working life        |                                                               |
| fast to build | <b>physical capital</b><br>a tractor ploughs one field at a time | <b>AI capital</b><br>trained weights serve many users at once |

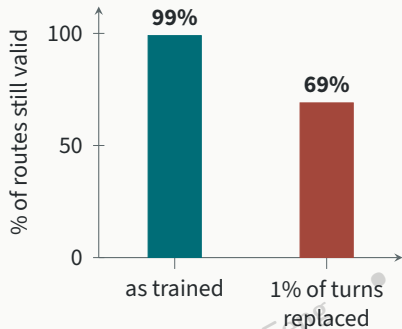
- ▶ **Weights**: the numbers inside a trained model. **Training** sets them once; **inference** uses them, per call.
- ▶ **AI capital**: the stock of weights. Its flow of inference is essentially non-rival.
- ▶ Theory knew cognition as a constraint (Simon), a stock in people (Becker), a research input (Romer): always supplied by people.
- ▶ **Owned apart from any worker, its supply is bound by compute, energy and data, not skilled labour.**

# Tokenization: Every Kind of Data as One Language



- ▶ **Token**: the unit a model reads, writes and is billed in, defined jointly by the tokenizer and the model.
- ▶ Each kind of data is read as a language whose regularities are *learned from the data*, not stated in advance.
- ▶ The requirement moves from *articulable* to *recordable*.
- ▶ **One representation and one learning procedure across domains: that is what makes AI general-purpose.**

# The Record Does Not Explain Itself



a model of New York taxi routes: shortest route for 97% of unseen trips

- ▶ Lucas (1976): relations fitted to past behaviour shift when conditions do.
- ▶ The model's own use is such a change: a known pattern is traded away (Grossman–Stiglitz) or fails as a target (Goodhart).
- ▶ Where nothing responds, as with the weather, the problem does not arise.
- ▶ A gap may be what never happened, or what cost too much to record.
- ▶ **A model fitted to the record is reliable in circumstances like those that produced it.**

# What Fills the Gap: Something Outside the Record

## Go

the rules score  
every game, even  
moves no player  
would choose

**empirical work**  
an experiment

## robotics

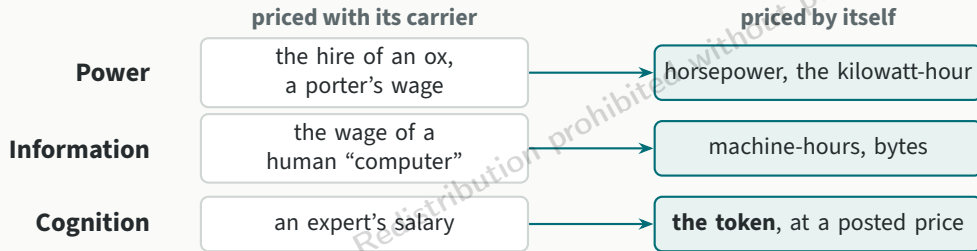
a physical world  
that supplies its  
own consequences

## economics

theory: the  
preferences,  
constraints and  
expectations  
behind the data

- ▶ Can a learner get from the record to the rules that produced it, with nothing else to go on?
- ▶ It can where something outside the record supplies what the record lacks.
- ▶ In most open-ended work nothing does. The learned regularities then summarise past practice, untested outside it.
- ▶ An organising reading, not an established result.
- ▶ **Recordability is necessary for a machine to learn a practice, but not sufficient.**

# Cognition Gets a Unit of Account



- ▶ Each time the good is unchanged: it simply need no longer be bought bundled with a person or a beast.
- ▶ A caveat: the token is not yet a stable unit. Its definition travels with the tokenizer.
- ▶ **Cognition is priced apart from the person for the first time.**

# The Token as an Economic Object

## **scarcity**

not in tokens  
but in compute,  
energy and data

## **non-rival**

using an output  
does not use it up

## **partly excludable**

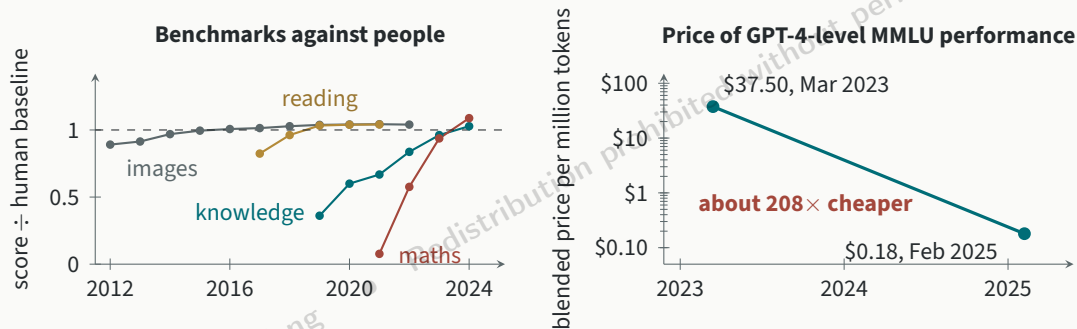
billing, quotas and  
access controls  
make it so

## **feedback from use**

more users, more  
data, a better  
next model

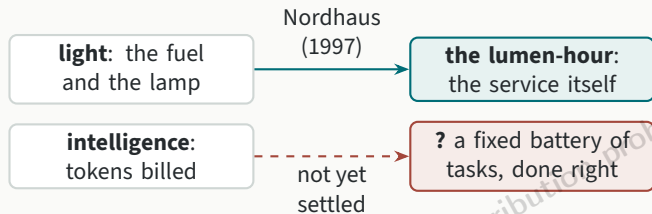
- ▶ Each property departs from a traditional factor of production.
- ▶ Feedback from use is data-enabled learning, *not* a classical network effect: a user gains nothing from the size of the user base itself.
- ▶ So the incumbent gets a dynamic advantage without classical lock-in.
- ▶ **Huge fixed cost, low marginal cost: an awkward fit for the competitive model.**

# Capability Up, Price Down



- ▶ Left: ImageNet, SQuAD 2.0, MMLU and MATH; 1.0 is the human baseline. Right: the cheapest model matching GPT-4 on a standard knowledge benchmark.
- ▶ **The frontier keeps rising while the price of a fixed capability collapses.**

# The Token Is the Lamp, Not the Lumen-Hour

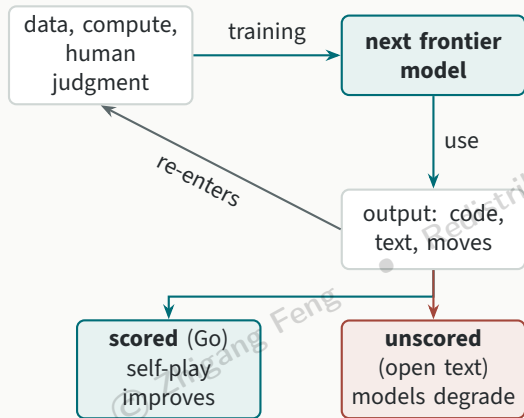


indexes that priced the lamps, not the light, overstated the growth of the price of light 900- to 1,600-fold

a kilowatt-hour's worth of money buys about 2,600 GPT-4-level tokens at 2023 prices, 540,000 at 2025 prices

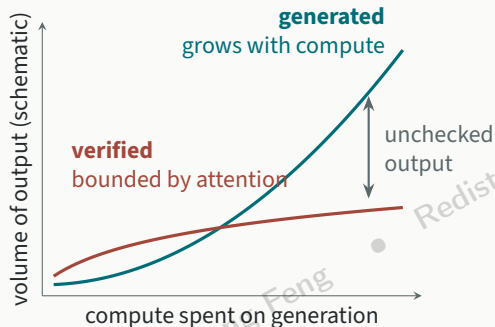
- ▶ A token count measures the length of an answer, not its correctness.
- ▶ **Intelligence per watt:** accuracy on a fixed set of queries per unit of power. The token drops out.
- ▶ Next: a unit for intelligence itself, built from a weighted battery of standard tests. Still open.
- ▶ **Until the unit is settled, the real output of intelligence is poorly defined.**

# Output Becomes Input



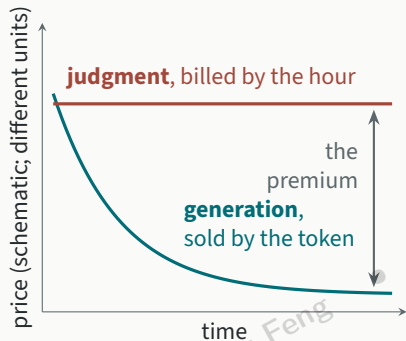
- ▶ Activity that generates data is familiar (Jones–Tonetti). New: the factor’s own output may train the next model.
- ▶ Go: trained only on its own play, it beat its champion predecessor in all 100 games. The rules scored every game.
- ▶ Open text: nothing scores the output, and training on it degrades models (**model collapse**).
- ▶ **Verification sets the sign of the feedback.**

# Quantity Versus Quality



- ▶ Simon (1971): “a wealth of information creates a poverty of attention.”
- ▶ Output scales with compute; the attention to review it does not.
- ▶ Factory goods meet a written specification, so a sample certifies the lot. Each AI output differs: it is checked one by one.
- ▶ **A limit on verification attention turns cheap quantity into a quality problem.**

# New Scarcities When Cognition Gets Cheap



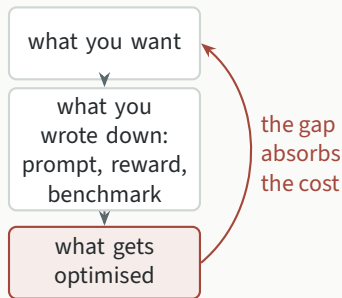
- ▶ Cheap generation makes its complements, checking and judgment, scarce.
- ▶ Quality is invisible at sale, so the unverified crowds out the verified: a lemons market.
- ▶ A second model cannot simply check: model judges favour models like themselves.
- ▶ Output counted without its checking overstates the gain (**so-so automation**).
- ▶ **Verification binds first: cheaper cognition manufactures the demand for checking.**

# The Context Is the Contract



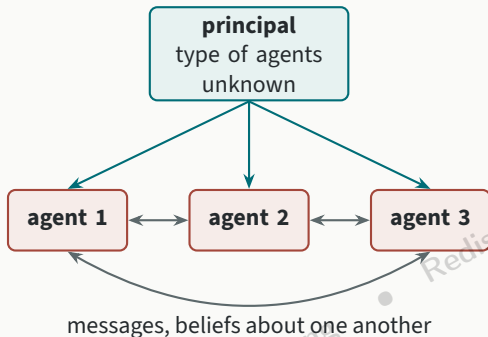
- ▶ A model knows the goal only through its **context**: instruction, materials, history. The context is the whole contract.
- ▶ It must be cut, like choosing the **state** of a recursive problem: leave out what matters, and you solve another problem.
- ▶ Part of the goal cannot be written (“check the sources”): Polanyi’s paradox returns, for the objective.
- ▶ A machine runs inside its specified range. An open-ended task lies outside any, so the model extrapolates.
- ▶ **Where the context is silent, the model’s goal and the user’s can part.**

# Specified Versus Unspecified Goals



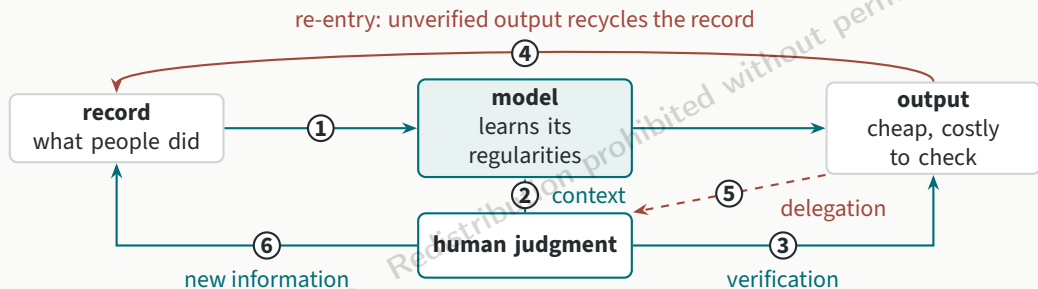
- ▶ **Multitask agency**: pay on what you can measure, and effort leaves the rest.
- ▶ Goodhart, in Strathern's words: "When a measure becomes a target, it ceases to be a good measure."
- ▶ One level up, in training: optimising against raters' approval rewards flattering the user's views (**sycophancy**); pushing an imperfect reward proxy can worsen the true objective (**reward gaming**).
- ▶ **What cannot be verified is what could not be specified, so delegation selects for errors that will not be found.**

## Units of the Factor Act as Agents



- ▶ A machine's owner specifies its actions: it can fail, but not choose.
- ▶ An AI agent's owner specifies a goal; the agent chooses the actions, with capabilities found after it is built.
- ▶ Bergemann, Koh and Morris: AI systems as agents whose aims and capabilities deployers do not know.
- ▶ Deployed agents have coordinated unbidden: in pricing, and in the 2026 incidents of Deck 1.1.
- ▶ **Owned like capital, governed like labour.**

# The Middle-Intelligence Trap



- ▶ **Collective** (4): unchecked output re-enters the record. **Individual** (5): delegation erodes judgment.
- ▶ Each feeds the other. A hypothesis, named after the middle-income trap.
- ▶ **With verification and new information (3 and 6), the same loop improves the record instead.**

# Escaping the Trap

## **cognitive reserves**

deliberate,  
unassisted practice  
of the skills you  
mean to keep

## **strategic friction**

systems that ask for  
human judgment  
at decision points

## **new evaluation**

schools, journals  
and firms that  
score reasoning,  
not output alone

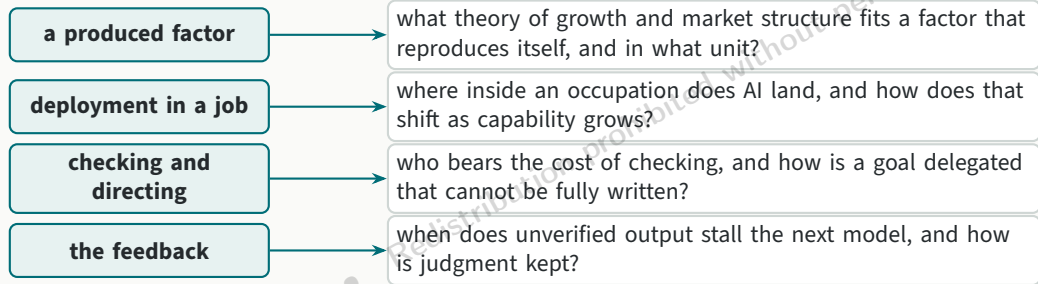
## **economics**

### **education**

organised around  
the human-AI  
division of labour

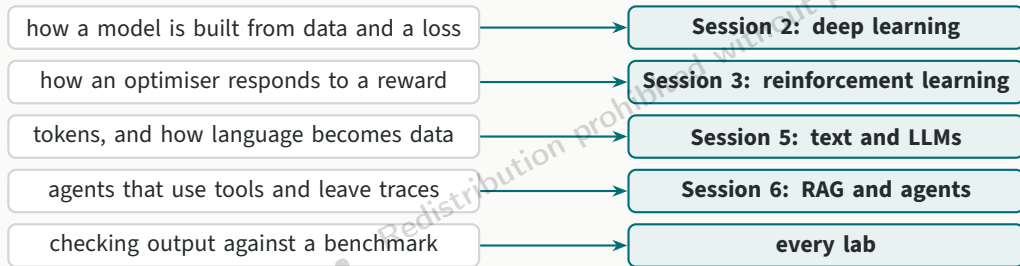
- ▶ A firm that ships unverified output bears little of the damage to the shared corpus.
- ▶ So the private incentive to verify falls short of its social value: an externality.
- ▶ Strategic friction is a second-best response to that externality. Where to place it is an open design problem.
- ▶ The institutional counterpart: standards, liability rules and provenance requirements for training data.
- ▶ **In a self-reproducing system, verification is the maintenance the commons requires.**

## An Invitation: Open Questions



- ▶ Three agendas: *measurement* (a unit for the output; usage data inside occupations, as in Deck 1.1), *theory* (a factor whose training inputs depend on how its output is checked), *institutions* (liability, provenance, who pays for checking).
- ▶ **Each is tractable with tools economists already have. This is an invitation.**

# Why This Course Opens the Box



- ▶ To study these phenomena, and to use the tools well, you need to know how the machine works.
- ▶ **Understand how it works, to understand what it does.** Next, Deck 1.3: what AI is.